Algorithm 12: Bi-directional Stitcher

Input:

- V. Set of nodes
- Labels_F. Map of Pareto-optimal forward labels from Algorithm 10
- Labels_B. Map of Pareto-optimal backward labels from Algorithm 11
- Record. The global static record to beat (e.g., 13)
- MinWait. Minimum connection time required at a station (e.g., 5 or 10 mins)

Output:

- P_{best}. The reconstructed sequence of labels representing the best route, or null if no route beats the record.

best_route $\leftarrow$ null
best_countries $\leftarrow \emptyset$
best_duration $\leftarrow \infty$
// 1. Scan the intersection of nodes
for all $v \in V$ **do**
    if $\text{Labels}_F(v) = \emptyset \vee \text{Labels}_B(v) = \emptyset$ **then**
        continue // Node was not reached by both halves
    // 2. Iterate through all combinations of forward and backward labels at node 'v'
    for all $L_f \in \text{Labels}_F(v)$ **do**
        for all $L_b \in \text{Labels}_B(v)$ **do**
            // — CHECK A: TIME CONTINUITY —
            // Does the forward path arrive in time to catch the backward path's departure?
            wait_time $\leftarrow L_b.t - L_f.t$
            if wait_time $<$ MinWait **then**
                continue
            // — CHECK B: 24-HOUR CONSTRAINT —
            // Is the total combined elapsed time strictly within 24 hours?
            total_duration $\leftarrow L_b$.end_time $- L_f$.start_time
            if total_duration > 1440 **then**
                continue
            // — CHECK C: GEOGRAPHIC UNION —
            // Merge the two sets of countries (this naturally deduplicates visited countries)
            $C_{\text{combined}} \leftarrow L_f.C \cup L_b.C$
            // — EVALUATE AND UPDATE —
            // We strictly want to beat the record. If it ties the current best, we can tie-break by duration.
            if $|C_{\text{combined}}| >$ Record $\vee$ ($|C_{\text{combined}}| =$ Record $\wedge$ total_duration $<$ best_duration) **then**
                Record $\leftarrow |C_{\text{combined}}|$ // Dynamically raise the bar for the rest of the loop
                best_countries $\leftarrow C_{\text{combined}}$
                best_duration $\leftarrow$ total_duration
                // 3. Reconstruct the stitched path
                best_route $\leftarrow$ ReconstructStitchedPath(L_f, L_b)
return best_route
// Auxiliary Function: ReconstructStitchedPath(L_f, L_b)

$P_{\text{forward}} \leftarrow$ empty sequence
curr $\leftarrow L_f$
// Trace forward path backwards to the start

while curr $\neq$ null **do**
    prepend curr to P_{forward}
    curr $\leftarrow$ curr.pred
$P_{\text{backward}} \leftarrow$ empty sequence
// Note: We skip the first node of L_b to avoid duplicating the meeting station 'v'
curr $\leftarrow L_b$.succ
// Trace backward path forwards to the destination
while curr $\neq$ null **do**
    append curr to P_{backward}
    curr $\leftarrow$ curr.succ
return P_{forward} concatenated with P_{backward}